

VisCaption

Image-to-Text Generation

Using CLIP Vision Encoders and GPT-2 Language Decoders

IE 7615 Deep Learning for AI

Instructor: Prof. Xuemin Jin

Group 8: Quoc Hung Le, Khoa Tran, Hassan Alfaree

COCO 2017

118 K

Training Images

500

Val Images Evaluated

3

Fine-tuning Strategies

0.9364

Best CIDEr Score

Presentation Outline

01

Problem & Motivation

Why image captioning matters

02

Architecture

CLIP + ProjectionHead + GPT-2

03

Fine-Tuning Strategies

Frozen vs FT-4L vs LoRA

04

Results

COCO 500-image benchmark

05

Sensitivity Analysis

Beam width and temperature

06

Key Findings

What the numbers tell us

07

Ethical Considerations

Bias, risk, and responsibility

Why Image Captioning?

2.2B

visually impaired people
globally need alt-text
([WHO, 2026](#))

10X

faster content indexing
with auto-generated metadata
([MomentsLab, 2026](#))

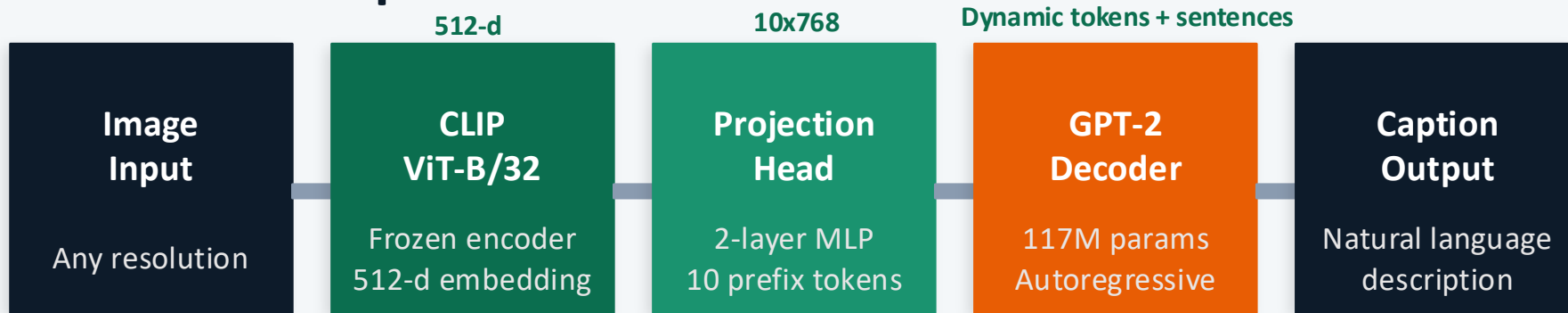
55.5%

of web images lack
accessible descriptions
([wearetenet, 2026](#))

The Task

Given an arbitrary image, generate a fluent, accurate natural-language description of its content - without any human annotation at inference time.

End-to-End Pipeline



ProjectionHead Detail



Image prefix conditioning: visual context is injected as 10 learnable prefix tokens prepended to GPT-2's input.

Three Training Strategies Compared

Frozen

62.9M params

Only ProjectionHead is trained.
GPT-2 weights stay completely fixed.
Fastest to train, smallest memory footprint.

Baseline

FT-4L

129.9M params

ProjectionHead + last 4 GPT-2
transformer blocks fine-tuned.
Highest raw capacity.

Best Accuracy

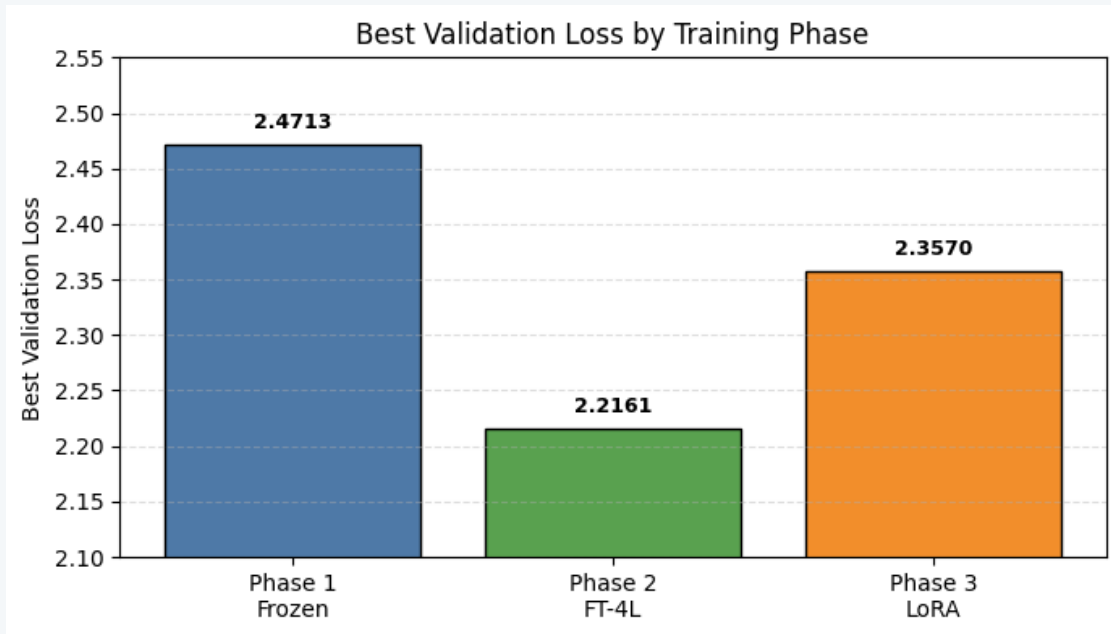
LoRA $r=8, a=16$

63.76M params

Low-rank adapters injected
into GPT-2 attention layers via PEFT.
 $\alpha=16$, scaling factor = 2.0.

Most Efficient

Split COCO 2017 (106K train, ~6K val, ~6K test)



Phase 1: Frozen

20 epochs

Early stopped, patience=3

Phase 2: FT-4L

15 epochs

Converges faster with LR=5e-5

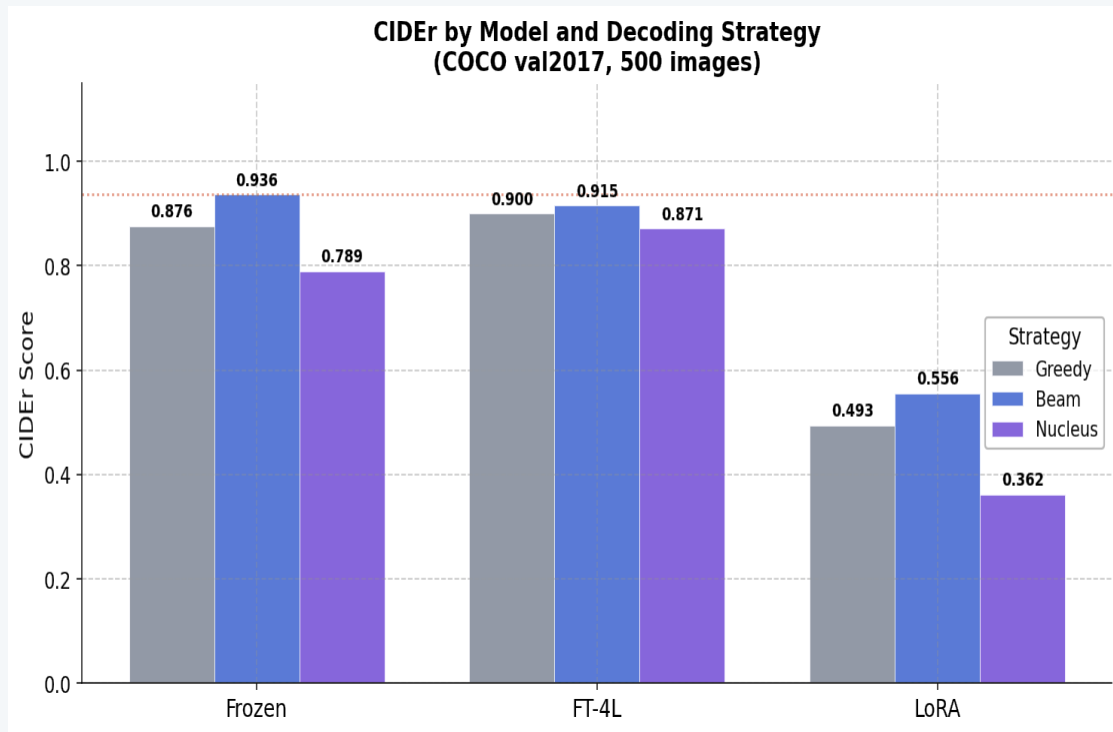
Phase 3: LoRA ($r=8$, scale =2)

15 epochs

Should change params

04 RESULTS

COCO val2017: 500 Images - All Configurations



Change LoRA r=32, scale =2; Evaluate on 5000 images

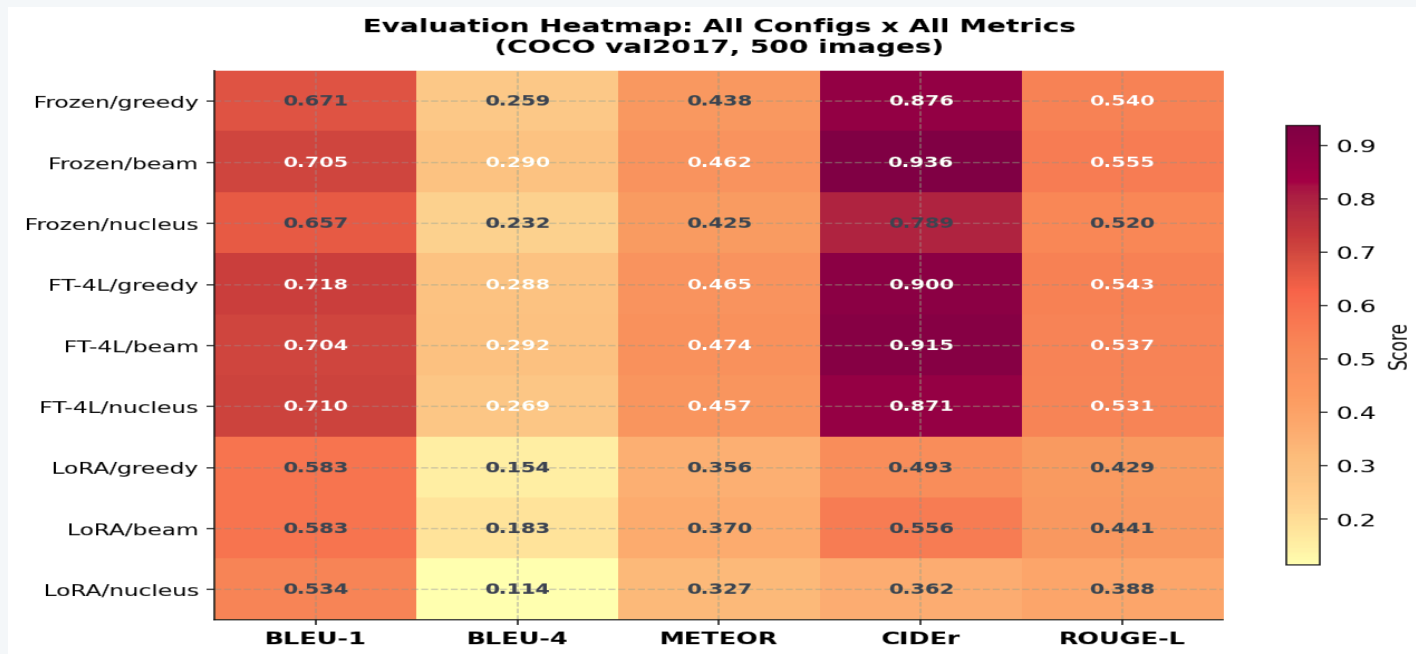
Best per Model (Beam Search)

Model	CIDEr	BLEU-4
Frozen	0.9364	0.2896
FT-4L	0.9364	0.2917
LoRA	0.5556	0.1835

Key Insight

Frozen/Beam = FT-4L/Beam on CIDEr
despite 67M fewer parameters

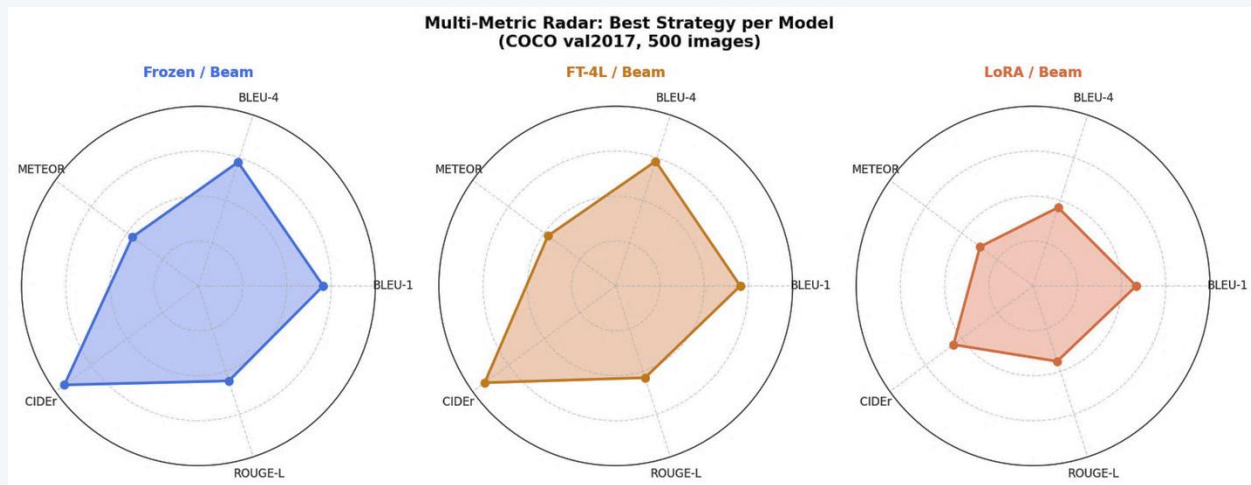
Complete Evaluation Heatmap: 9 Configurations x 5 Metrics



Warmer colors = higher score within each metric column.

Best row: FT-4L / Beam (highlighted). Evaluation on COCO val2017, 500 images.

Multi-Metric Radar: Best Strategy per Model



Frozen / Beam

Competitive on all metrics
with zero LM fine-tuning

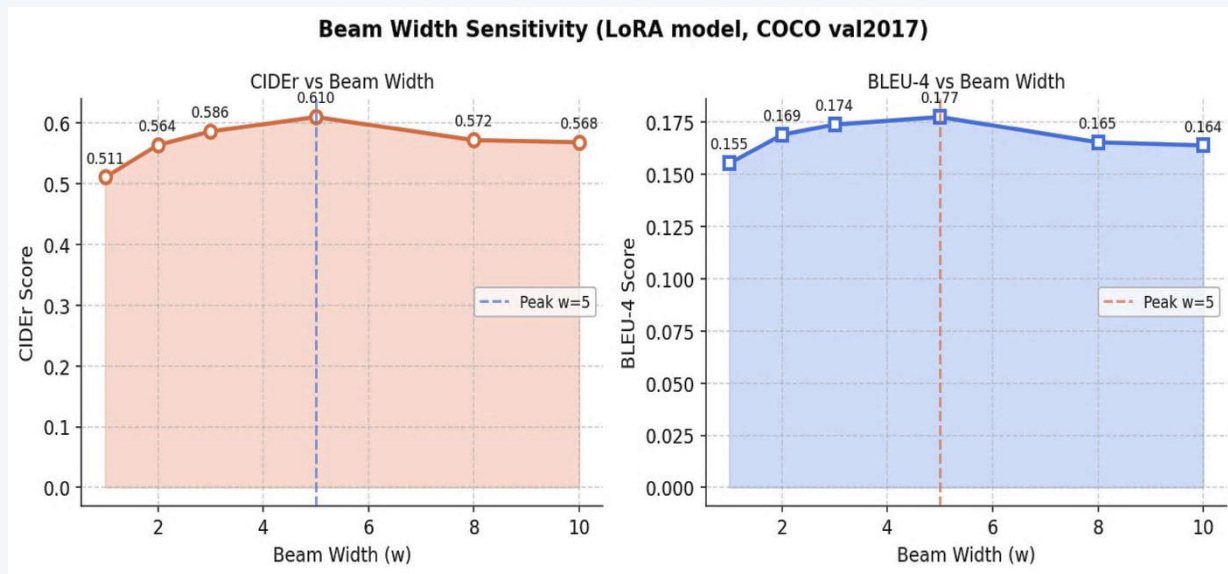
FT-4L / Beam

Best METEOR (0.474)
Marginally better than
Frozen

LoRA / Beam

Lower scores on COCO scale
Parametric efficiency trade-
off -> upgrade by r=32

Beam Width Sweep: CIDEr and BLEU-4

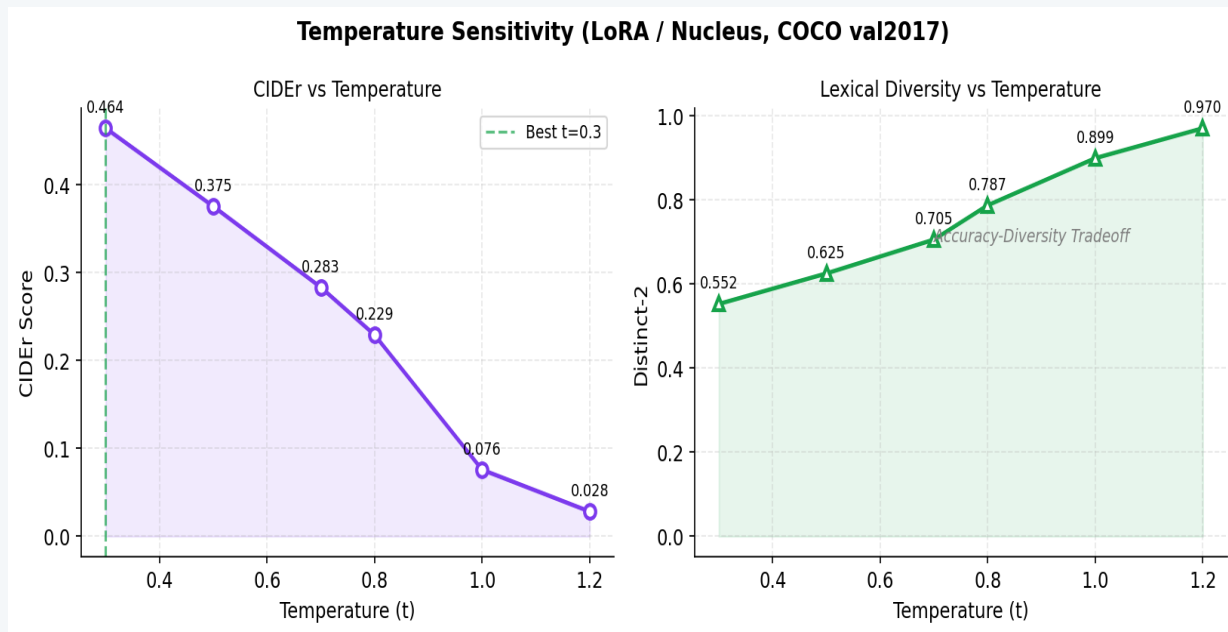


Beam Sweep Results

w=1	0.5109	0.1553
w=2	0.5638	0.1690
w=3	0.5861	0.1739
w=5	0.6099	0.1775
w=8	0.5718	0.1654
w=10	0.5684	0.1639

Recommended: w = 5

Temperature Sweep: Accuracy vs. Diversity Trade-off



Accuracy - Diversity

t=0.3	0.464	0.552
t=0.5	0.375	0.625
t=0.7	0.283	0.705
t=0.8	0.229	0.787
t=1.0	0.076	0.899
t=1.2	0.028	0.970

CIDEr / Distinct-2

Use t=0.3 for accuracy, t=0.5 for diversity

06 FINDINGS

01

Prefix conditioning works without LM fine-tuning

CIDEr 0.9364

Frozen/Beam matches FT-4L/Beam exactly. ProjectionHead alone provides sufficient visual conditioning on 118K images.

02

Beam search wins on all accuracy metrics

+12% vs Greedy

Beam search outperforms greedy and nucleus across all 3 models. Gap is largest for LoRA (+12.4% CIDEr).

03

Optimal beam width is dataset-dependent

w=5 on COCO

w=5 peaks on COCO vs. w=8 on Flickr30k. Tune on a held-out validation set, not fixed heuristics.

04

Temperature degrades accuracy monotonically

CIDEr:
0.46→0.03

CIDEr drops from 0.46 at t=0.3 to 0.03 at t=1.2. Diversity rises proportionally. Use t=0.3 for factual captioning.

Ethical Considerations

Dataset Bias

COCO was annotated by workers with Western, English-speaking demographics:

- Cultural bias
- Linguistic bias

→ Train on diverse multilingual datasets; evaluate across demographics

Misrepresentation Risk

Models generate confident captions even for out-of-distribution images. In accessibility tools (screen reader alt-text), incorrect captions actively mislead visually impaired users → Deploy with human-in-loop for high-impact accessibility applications

Accessibility Value

CIDEr 0.93 on common COCO scenes approaches usable quality for assistive technology. 2.2B visually impaired people could benefit from automatic alt-text generation when deployed responsibly → Human review pipeline; confidence thresholds; grounding supervision

Privacy & Surveillance

Automatic scene description at scale could enable mass surveillance by generating textual logs from camera feeds → Consent frameworks; on-device processing; data minimization

8. DEMO

VisCaption: Interactive Web Dashboard

Live Inference (Both CLI and Web)

Upload any image, select model and strategy, generate captions in real time with preprocessing pipeline visualization.

Dynamic Metrics

No Ground-Truth panel: CLIPScore, Perplexity, Grammar Score computed instantly. GT References panel: BLEU/METEOR/CIDEr.

Sensitivity Analysis

Real-time 12-inference sweep on your image. Static COCO 500 charts for beam width and temperature.

Analysis Page

Key findings and ethical considerations grounded in COCO 500 evaluation data.

LAUNCH DEMO

```
1. Web app:  
bash run.sh  
2. CLI  
demo.ipynb
```

Go ahead....

8. DEMO

Web demo

Live Inference | CLIP ViT-B/32 + GPT-2

COCO 2017 – 118K

MPS / M1

Drop image or click to browse

Any resolution, CLIP center-crop normalization applied

JPG

PNG

WebP

BMP

TIFF

HEIC

GIF

ORIGINAL

500x375

CLIP-READY

224x224 preview



ORIGINAL
500x375

CROP POS.
37,0

FORMAT
JPG

STATUS
Normal

IMG

RGB

SZ

CRP

NRM

EMB

Decode > Conv > Resize > Crop > Norm > Embed

JPG

RGB

299x224

224

CLIP u

512-d

MODEL VARIANT

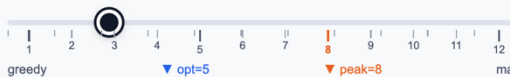
FT-4L: Fine-tuned 4 layers

DECODING STRATEGY

Beam Search

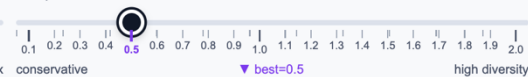
BEAM WIDTH

Width



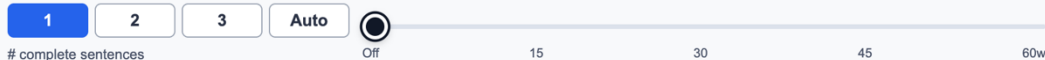
TEMPERATURE

Temp



SENTENCES

MAX WORDS OFF



Single Model

All 3 Models

Regenerate

CLIP: 447ms GPT-2: 1459ms

Total: 1906ms

FT-4L

beam

CLIPScore: 32.8

"a woman walking down the street at night."

8. DEMO

Web demo

Live Inference | CLIP ViT-B/32 + GPT-2

COCO 2017 -- 118K

MPS / M1

Drop image or click to browse

Any resolution, CLIP center-crop normalization applied

JPG

PNG

WebP

BMP

TIFF

HEIC

GIF

ORIGINAL

500x400



CLIP-READY

224x224 preview



ORIGINAL
500x400

CROP POS.
28,0

FORMAT
JPG

STATUS
Normal

IMG

RGB

SZ

CRP

NRM

EMB

Decode > Conv > Resize > Crop > Norm > Embed
JPG RGB 280x224 224 CLIP u 512-d

MODEL VARIANT

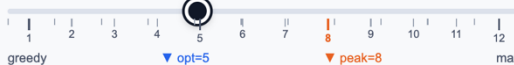
FT-4L: Fine-tuned 4 layers

DECODING STRATEGY

Beam Search

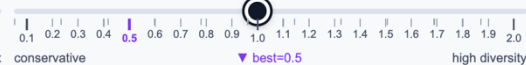
BEAM WIDTH

Width



TEMPERATURE

5 Temp



SENTENCES

MAX WORDS OFF

1

2

3

Auto

complete sentences

Off

15

30

45

60w

Single Model

All 3 Models

Regenerate

CLIP: 77ms GPT-2: 1272ms

Total: 1349ms

FT-4L beam

CLIPScore: 20.7

"a group of people that are in a room."

8. DEMO

Web demo

Live Inference | CLIP ViT-B/32 + GPT-2

COCO 2017 -- 118K

MPS / M1

Drop image or click to browse

Any resolution, CLIP center-crop normalization applied

JPG

PNG

WebP

BMP

TIFF

HEIC

GIF

ORIGINAL 500x333



CLIP-READY 224x224 preview



ORIGINAL
500x333

CROP POS.
56,0

FORMAT
JPG

STATUS
Normal

IMG

RGB

SZ

CRP

NRM

EMB

Decode > Conv > Resize > Crop > Norm > Embed

JPG

RGB

336x224

224

CLIP u

512-d

MODEL VARIANT

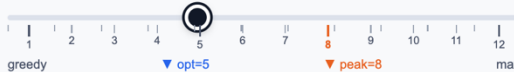
FT-4L: Fine-tuned 4 layers

DECODING STRATEGY

Beam Search

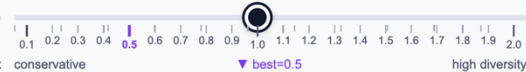
BEAM WIDTH

Width



TEMPERATURE

Temp



SENTENCES

MAX WORDS OFF

1

2

3

Auto

complete sentences

Off

15

30

45

60w

Single Model

All 3 Models

Regenerate

CLIP: 79ms GPT-2: 1228ms

Total: 1307ms

FT-4L

beam

CLIPScore: 28.9

"a man riding a wave on top of a surfboard."

8. DEMO

Web demo

Live Inference | CLIP ViT-B/32 + GPT-2

COCO 2017 -- 118K

MPS / M1

Drop image or click to browse

Any resolution, CLIP center-crop normalization applied

JPG

PNG

WebP

BMP

TIFF

HEIC

GIF

ORIGINAL 500x333



CLIP-READY 224x224 preview



ORIGINAL
500x333

CROP POS.
56,0

FORMAT
JPG

STATUS
Normal

IMG

RGB

SZ

CRP

NRM

EMB

Decode > Conv > Resize > Crop > Norm > Embed

JPG

RGB

336x224

224

CLIP u

512-d

MODEL VARIANT

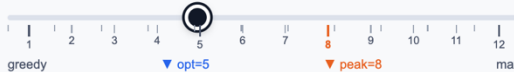
FT-4L: Fine-tuned 4 layers

DECODING STRATEGY

Beam Search

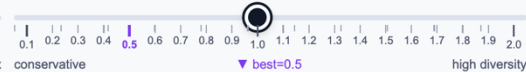
BEAM WIDTH

Width



TEMPERATURE

Temp



SENTENCES

MAX WORDS OFF

1

2

3

Auto

complete sentences

Off

15

30

45

60w

Single Model

All 3 Models

Regenerate

CLIP: 79ms GPT-2: 1228ms

Total: 1307ms

FT-4L

beam

CLIPScore: 28.9

"a man riding a wave on top of a surfboard."

8. DEMO

Web demo

Evaluation Metrics

Dynamic

Static (COCO 500)

Test-Set Aggregate Metrics

No Ground-Truth

Ground-Truth References

Reference-free metrics computed directly from the generated caption and the uploaded image. Generate captions in Live Demo first.

CLIPScore

cosine_similarity(CLIP_image, CLIP_caption)

28.70

Range 0-100: human threshold ~67

CLIP Text Consistency

Avg cosine sim among 3 caption paraphrases via CLIP text encoder

0.98

Range 0-1: higher = more internally consistent

Perplexity

Language model fluency: lower = more fluent

34.4

Typical good captions: 20-80: lower is better

Grammar Score

Error-free ratio via rule-based grammar check

100%

% of tokens with no grammar errors detected

Distinct-2

Unique bigrams / total bigrams across 3 captions

0.6552

Higher is better (above 0.7 = diverse)

Self-BLEU

Avg BLEU of each caption vs others (3 captions)

0.7363

Lower is better (below 0.35 = diverse)

Computed via /api/diversity_metrics | 19 unique bigrams / 29 total

Compute Reference-Free Metrics

Done

8. DEMO

Web demo

Evaluation Metrics

Dynamic

Static (COCO 500)



Live inference result

410

CLIP MS

3840

GPT-2 MS

4250

TOTAL MS

Normal

IMAGE TYPE

FROZEN / BEAM

"a surfer riding a wave on a surfboard."

FT-4L / BEAM

"a man riding a wave on top of a surfboard."

LORA / BEAM

"a man on a surfboard riding a wave in the ocean."

Test-Set Aggregate Metrics

No Ground-Truth

Ground-Truth References

GROUND-TRUTH REFERENCES (FOR THIS IMAGE)

a man riding a surfboard on the wave in the ocean

Calculate BLEU / METEOR / CIDEr / ROUGE-L

Done — 3 captions scored

Per-Caption Scores vs 1 reference(s)

MODEL/STRATEGY	BLEU-1	BLEU-4	METEOR	ROUGE-L	CAPTION
Frozen / beam	0.5338	0.0636	0.3948	0.4211	"a surfer riding a wave on a surfboard."
FT-4L / beam	0.6364	0.2900	0.5204	0.4762	"a man riding a wave on top of a surfboard."
LoRA / beam	0.8333	0.3170	0.8036	0.7273	"a man on a surfboard riding a wave in the ocean."

8. DEMO

Web demo

Parameter Sensitivity

Beam width sweep • Temperature sweep

Dynamic

Static (COCO 500)



REAL-TIME PARAMETER SWEEP (CURRENT IMAGE)

Model

References (optional)

FT-4L

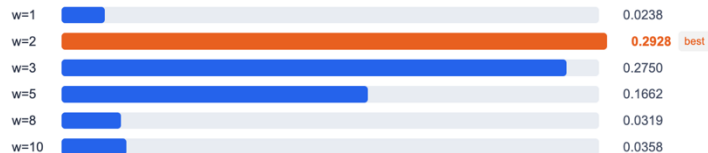
a man riding a surfboard on the wave in the ocea

Run Sweep

Done — 6 beam + 6 temp inferences

Beam Width - FT-4L

strategy=beam | 6 runs: w=1,2,3,5,8,10



W	BLEU-4	METEOR	CAPTION
1	0.0238	0.3333	"a surfer riding a wave on top of a surfboard. . . .a ...
2	0.2928	0.4935	"a man riding a surfboard on top of a wave. pic.twitter...
3	0.2750	0.4893	"a man riding a surfboard on top of a wave. pic.twitter...
5	0.1662	0.4893	"a man riding a wave on top of a surfboard. pic.twitter...
8	0.0319	0.3583	"a surfer riding a wave on top of a surfboard. pic.twitt...
10	0.0358	0.3644	"a surfer riding a wave on top of a surfboard. pic.twitt...

Temperature - FT-4L

strategy=nucleus | 6 runs: t=0.3,0.5,0.7,0.8,1.0,1.2



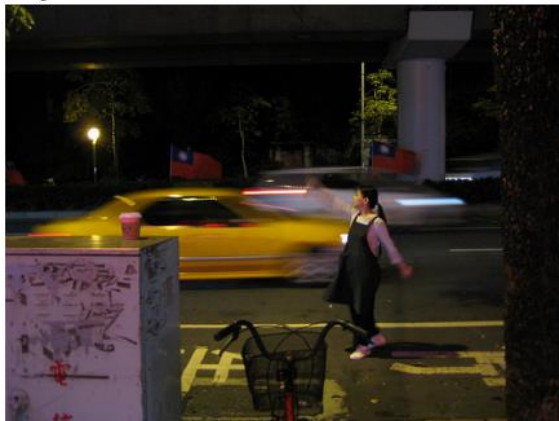
T	BLEU-4	METEOR	CAPTION
0.3	0.1508	0.5175	"a man riding a surfboard on top of some waves.
0.5	0.1656	0.6657	"a man riding a surfboard on top of some waves.
0.7	0.1929	0.4617	"a man riding a surfboard on top of a wave. pic .jpg
0.8	0.0149	0.2593	"a surfer riding a wave on top of his surfboard. . . .a...
1	0.0188	0.3606	"a surfer riding a big wave on his surfboard.ongsurfin...
1.2	0.0080	0.1515	"wistful surfer coming out of inverted swing stand. mi...

8. DEMO

CLI

Caption Gallery | Strategy: beam | 3 of 3 images

FT-4L:
a woman walking down the street
at night.



FT-4L:
a group of people that are in a
room.



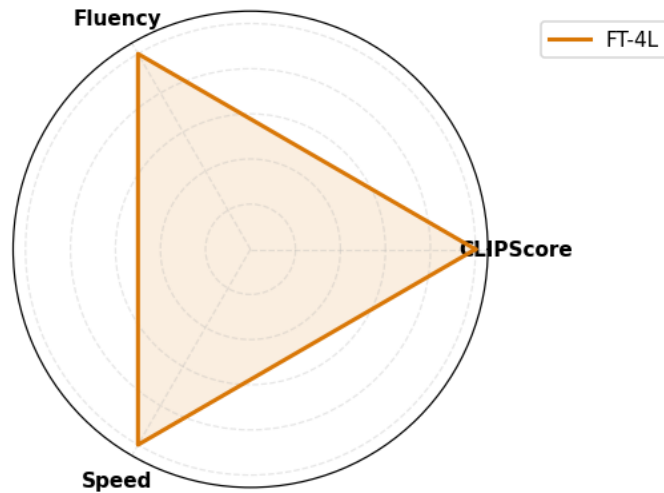
FT-4L:
a man riding a surfboard on top
of a wave.



8. DEMO

CLI

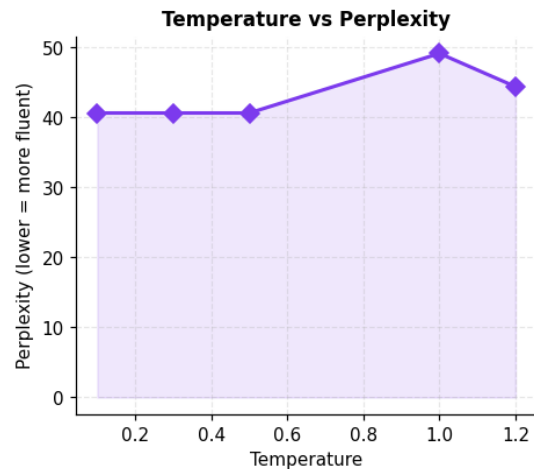
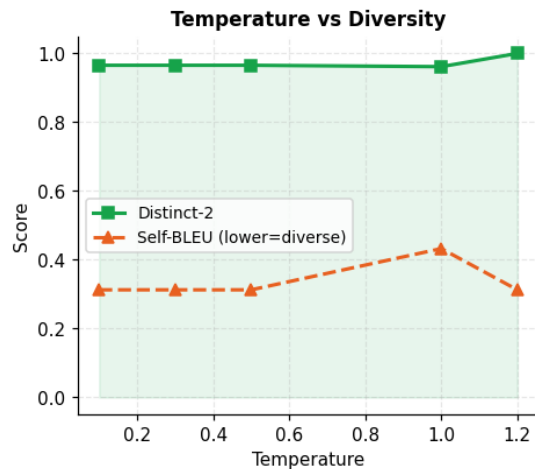
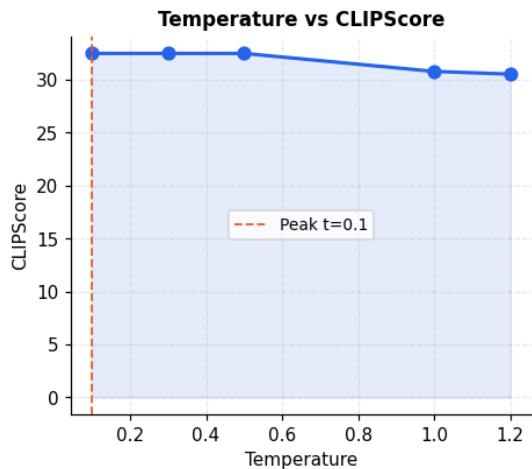
Model Radar | strategy: beam | normalized per metric



8. DEMO

CLI

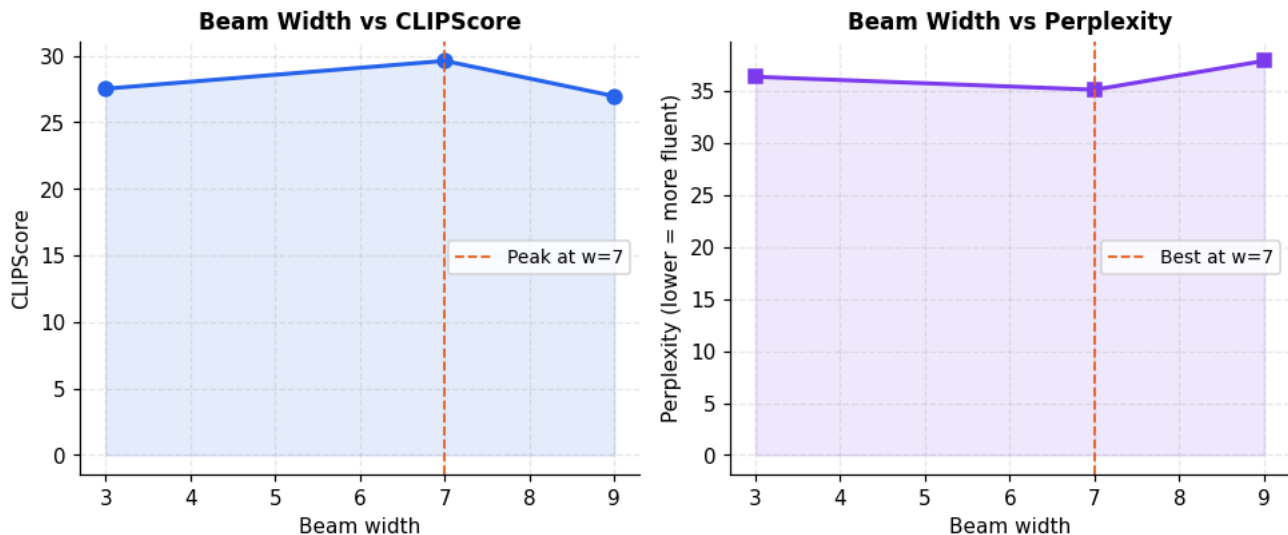
Temperature Sensitivity | model: FT-4L | 3 images



8. DEMO

CLI

Beam Width Sensitivity | model: FT-4L | 3 images



Conclusion

Best result: FT-4L / Beam achieves CIDEr 0.9364 on COCO val2017 (500 images). Frozen / Beam ties exactly with 67M fewer parameters, validating prefix conditioning as an effective approach.

Future Directions

Cross-attention conditioning to overcome the 10-token prefix bottleneck -> map GPT-2, like BLIP-2

Larger CLIP encoder (ViT-L/14, 768-d) for richer visual representation

BLIP-2 (3.1B, Q-Former) / GIT (680M, Unified Transformer) comparison